

Lecture 19: Orthogonalization (Second part)

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[Based on slides for „Coding the Matrix“]



WEEK

TEN



Agenda

Chapter 9

Lecture 19:

Orthogonal complement
Orthonormalization
QR factorization
solving $Ax = b$
Least square regression





Lecture 18 - Summary

Previous

Fire Engine problem vector to line

Concept & Definition

Generalization of fire engine problem to higher space

Theorems & Lemmas

Projection orthogonal from a vector to a list of vectors

Mutually orthogonal

orthogonalization

Tools

Project_orthogonal

orthogonalize

Find_basis

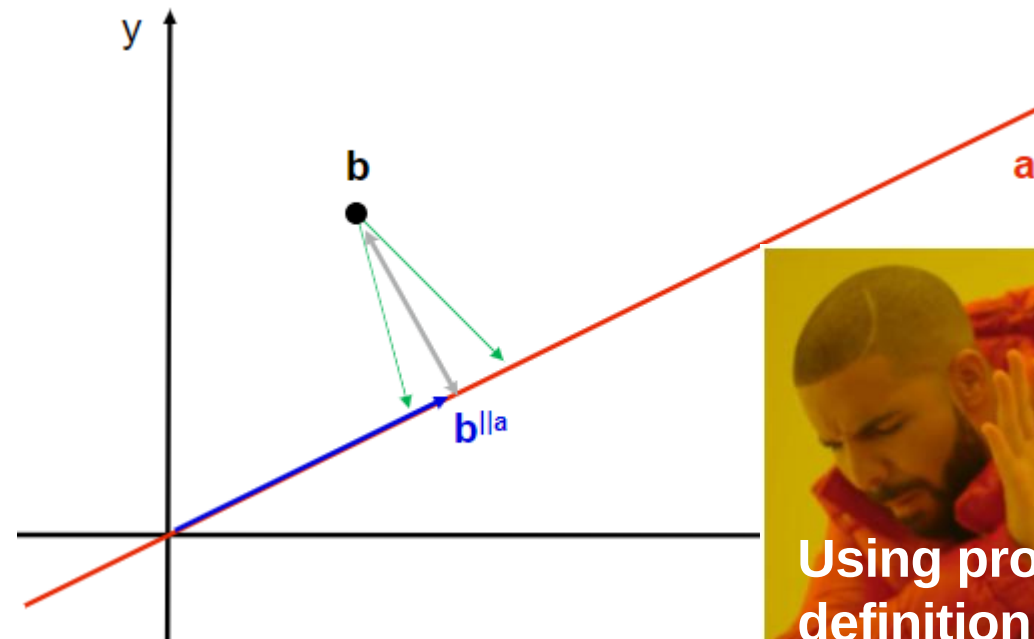
Find_subset_basis



Lecture 19(+) - Goal

Previous 1D Fire Engine Problem

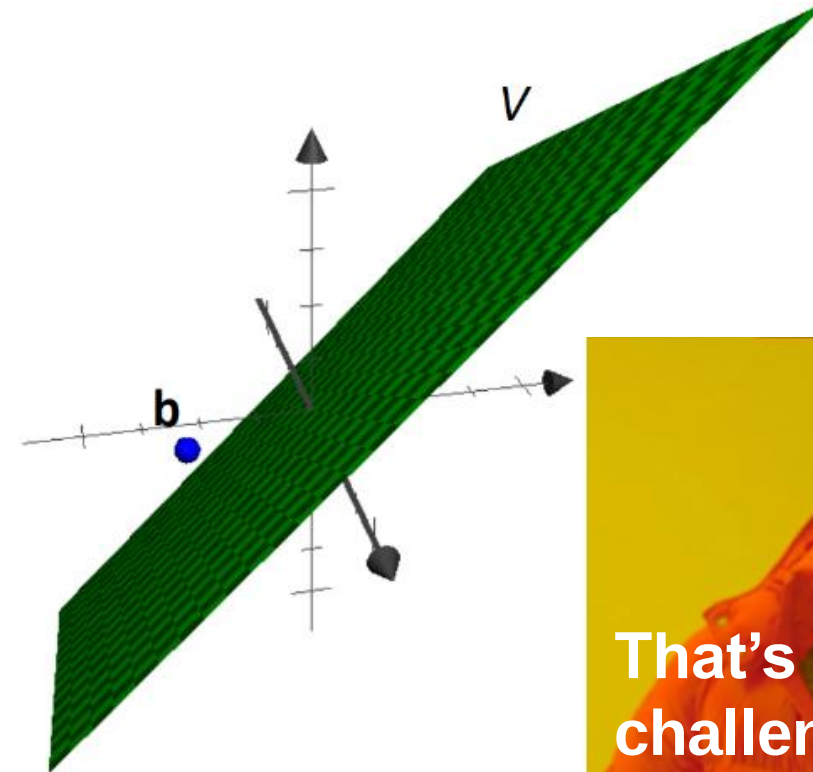
Previous goal: Find the closest point to $\mathbf{b}$ on the line $\mathbf{a}$.



Using projection
definition 8.3.6
and Lemma 8.3.11
too easy for me

Generalization of Fire Engine Problem

New goal: Find the closest point to $\mathbf{b}$ on the **plan** $\mathbf{V}$.



Now consider that the plan $\mathbf{V}$ is build by a list of vectors
 $\text{Span} \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$

That's a real
challenge for me
!



Orthogonal Complement

Let U be a subspace of W .

For each vector $\mathbf{b}$ in W ,

Let V be the set $\{\mathbf{b}^{\perp U} : \mathbf{b} \in W\}$.

Definition: We call V the *orthogonal complement* of U in W .

Easy observations:

- ▶ Every vector in V is orthogonal to every vector in U .
- ▶ Every vector $\mathbf{b}$ in W can be written as the sum of a vector in U and a vector in V .

Maybe $\mathcal{U} \oplus \mathcal{V} = \mathcal{W}$? To show direct sum of $\mathcal{U}$ and $\mathcal{V}$ is defined, we need to show that the only vector that is in both $\mathcal{U}$ and $\mathcal{V}$ is the zero vector.

Any vector $\mathbf{w}$ in both $\mathcal{U}$ and $\mathcal{V}$ is orthogonal to itself.

Thus $0 = \langle \mathbf{w}, \mathbf{w} \rangle = \|\mathbf{w}\|^2$.

By Property N2 of norms, that means $\mathbf{w} = \mathbf{0}$.

Therefore $\mathcal{U} \oplus \mathcal{V} = \mathcal{W}$. **Recall:** $\dim \mathcal{U} + \dim \mathcal{V} = \dim \mathcal{U} \oplus \mathcal{V}$



Orthogonal Complement: Example 1

Example: Let $\mathcal{U} = \text{Span} \{[1, 1, 0, 0], [0, 0, 1, 1]\}$. Let $\mathcal{V}$ denote the orthogonal complement of $\mathcal{U}$ in $\mathbb{R}^4$. What vectors form a basis for $\mathcal{V}$?

Every vector in $\mathcal{U}$ has the form $[a, a, b, b]$.

Therefore any vector of the form $[c, -c, d, -d]$ is orthogonal to every vector in $\mathcal{U}$.

Every vector in $\text{Span} \{[1, -1, 0, 0], [0, 0, 1, -1]\}$ is orthogonal to every vector in $\mathcal{U}$
... so $\text{Span} \{[1, -1, 0, 0], [0, 0, 1, -1]\}$ is a subspace of $\mathcal{V}$, the orthogonal complement of $\mathcal{U}$ in $\mathbb{R}^4$.

Is it the whole thing?

$\mathcal{U} \oplus \mathcal{V} = \mathbb{R}^4$ so $\dim \mathcal{U} + \dim \mathcal{V} = 4$.

$\{[1, 1, 0, 0], [0, 0, 1, 1]\}$ is linearly independent so $\dim \mathcal{U} = 2$... so $\dim \mathcal{V} = 2$

$\{[1, -1, 0, 0], [0, 0, 1, -1]\}$ is linearly independent
so $\dim \text{Span} \{[1, -1, 0, 0], [0, 0, 1, -1]\}$ is also 2....
so $\text{Span} \{[1, -1, 0, 0], [0, 0, 1, -1]\} = \mathcal{V}$.



Orthogonal Complement: Example 2

Example: Find a basis for the null space of $A = \begin{bmatrix} 1 & 0 & 2 & 4 \\ 0 & 5 & 1 & 2 \\ 0 & 2 & 5 & 6 \end{bmatrix}$

Thus the null space of A equals the orthogonal complement of Row A in $\mathbb{R}^4$.

Since the three rows of A are linearly independent, we know $\dim \text{Row } A = 3...$

so the dimension of the orthogonal complement of Row A in $\mathbb{R}^4$ is $4 - 3 = 1....$

The vector $[1, \frac{1}{10}, \frac{13}{20}, \frac{-23}{40}]$ has a dot-product of zero with every row of $A...$

so this vector forms a basis for the orthogonal complement.

and thus a basis for the null space of A .



Orthogonalization helps to find intersect of geometric object

Example: Find the intersection of

- ▶ the plane spanned by $[1, 2, -2]$ and $[0, 1, 1]$
- ▶ the plane spanned by $[1, 0, 0]$ and $[0, 1, -1]$



The orthogonal complement in $\mathbb{R}^3$ of the first plane is $\text{Span} \{[4, -1, 1]\}$.

Therefore first plane is $\{[x, y, z] \in \mathbb{R}^3 : [4, -1, 1] \cdot [x, y, z] = 0\}$

The orthogonal complement in $\mathbb{R}^3$ of the second plane is $\text{Span} \{[0, 1, 1]\}$.

Therefore second plane is $\{[x, y, z] \in \mathbb{R}^3 : [0, 1, 1] \cdot [x, y, z] = 0\}$

The intersection of these two sets is the set

$$\{[x, y, z] \in \mathbb{R}^3 : [4, -1, 1] \cdot [x, y, z] = 0 \text{ and } [0, 1, 1] \cdot [x, y, z] = 0\}$$

Since the annihilator of the annihilator is the original space, a basis for this vector

space is a basis for the null space of $A = \begin{bmatrix} 4 & -1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$

The null space of A is the orthogonal complement of $\text{Span} \{[4, -1, 1], [0, 1, 1]\}$ in $\mathbb{R}^3$... which is $\text{Span} \{[1, 2, -2]\}$



Computing the orthogonal complement

Suppose we have a basis $\mathbf{u}_1, \dots, \mathbf{u}_k$ for U and a basis $\mathbf{w}_1, \dots, \mathbf{w}_n$ for W . How can we compute a basis for the orthogonal complement of U in W ?

One way: use **orthogonalize(vlist)** with

$$\text{vlist} = [\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{w}_1, \dots, \mathbf{w}_n]$$

Write list returned as $[\mathbf{u}_1^*, \dots, \mathbf{u}_k^*, \mathbf{w}_1^*, \dots, \mathbf{w}_n^*]$

These span the same space as input vectors $\mathbf{u}_1, \dots, \mathbf{u}_k, \mathbf{w}_1, \dots, \mathbf{w}_n$, namely $\mathcal{W}$, which has dimension n .

Therefore exactly n of the output vectors $\mathbf{u}_1^*, \dots, \mathbf{u}_k^*, \mathbf{w}_1^*, \dots, \mathbf{w}_n^*$ are nonzero.

The vectors $\mathbf{u}_1^*, \dots, \mathbf{u}_k^*$ have same span as $\mathbf{u}_1, \dots, \mathbf{u}_k$ and are all nonzero since $\mathbf{u}_1, \dots, \mathbf{u}_k$ are linearly independent.

Therefore exactly $n - k$ of the remaining vectors $\mathbf{w}_1^*, \dots, \mathbf{w}_n^*$ are nonzero.

Every one of them is orthogonal to $\mathbf{u}_1, \dots, \mathbf{u}_n$...

so they are orthogonal to every vector in $\mathcal{U}$...

so they lie in the orthogonal complement of $\mathcal{U}$.

By Direct-Sum Dimension Lemma, orthogonal complement has dimension $n - k$, so the remaining nonzero vectors are a basis for the orthogonal complement.



Finding basis for the null space using the orthogonal complement

To find basis for null space of an $m \times n$ matrix $A = \begin{bmatrix} \mathbf{a}_1 \\ \vdots \\ \mathbf{a}_m \end{bmatrix}$,

find orthogonal complement of $\text{Span} \{\mathbf{a}_1, \dots, \mathbf{a}_m\}$ in $\mathbb{R}^n$:

- ▶ Let $\mathbf{e}_1, \dots, \mathbf{e}_n$ be the standard basis vectors $\mathbb{R}^n$.
- ▶ Let $[\mathbf{a}_1^*, \dots, \mathbf{a}_m^*, \mathbf{e}_1^*, \dots, \mathbf{e}_n^*] = \text{orthogonalize}([\mathbf{a}_1, \dots, \mathbf{a}_m, \mathbf{e}_1, \dots, \mathbf{e}_n])$
- ▶ Find the nonzero vectors among $\mathbf{e}_1^*, \dots, \mathbf{e}_n^*$



Towards QR factorization (1)

Matrix factorizations are useful mathematically and computationally:

- ▶ *Mathematical*: They provide insight into the nature of matrices—each factorization gives us a new way to think about a matrix.
- ▶ *Computational*: They give us ways to compute solutions to fundamental computational problems involving matrices.



Matrix with mutually orthogonal columns (1)

$$\begin{bmatrix} \mathbf{v}_1^{*T} \\ \vdots \\ \mathbf{v}_n^{*T} \end{bmatrix} \begin{bmatrix} | & & | \\ \mathbf{v}_1^* & \cdots & \mathbf{v}_n^* \\ | & & | \end{bmatrix} = \begin{bmatrix} \|\mathbf{v}_1\|^2 & & \\ & \ddots & \\ & & \|\mathbf{v}_n\|^2 \end{bmatrix}$$

Normalize columns

$$\begin{bmatrix} | & & | \\ \mathbf{v}_1^* & \cdots & \mathbf{v}_n^* \\ | & & | \end{bmatrix} \Rightarrow \begin{bmatrix} | & & | \\ \mathbf{q}_1 & \cdots & \mathbf{q}_n \\ | & & | \end{bmatrix}$$



Matrix with mutually orthogonal columns (2)

$$\begin{bmatrix} \mathbf{q}_1^T \\ \vdots \\ \mathbf{q}_n^T \end{bmatrix} \begin{bmatrix} \mathbf{q}_1 & \cdots & \mathbf{q}_n \end{bmatrix} = \begin{bmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{bmatrix}$$

Proposition: If columns of Q are mutually orthogonal with norm 1 then $Q^T Q$ is identity matrix.

Definition: Vectors that are mutually orthogonal and have norm 1 are *orthonormal*.

Definition: If columns of Q are orthonormal then we call Q a *column-orthogonal* matrix. *should be called orthonormal but oh well*

Definition: If Q is square and column-orthogonal, we call Q an *orthogonal* matrix.

Proposition: If Q is an orthogonal matrix then its inverse is Q^T .



Projection onto columns or a column orthogonal matrix

Suppose $\mathbf{q}_1, \dots, \mathbf{q}_n$ are orthonormal vectors.

Projection of $\mathbf{b}$ onto $\mathbf{q}_j$ is $\mathbf{b}^{\parallel \mathbf{q}_j} = \sigma_j \mathbf{q}_j$ where $\sigma_j = \frac{\langle \mathbf{q}_j, \mathbf{b} \rangle}{\langle \mathbf{q}_j, \mathbf{q}_j \rangle} = \langle \mathbf{q}_j, \mathbf{b} \rangle$

Vector $[\sigma_1, \dots, \sigma_n]$ can be written using dot-product definition of matrix-vector multiplication:

$$\begin{bmatrix} \sigma_1 \\ \vdots \\ \sigma_n \end{bmatrix} = \begin{bmatrix} \mathbf{q}_1 \cdot \mathbf{b} \\ \vdots \\ \mathbf{q}_n \cdot \mathbf{b} \end{bmatrix} = \begin{bmatrix} \frac{\mathbf{q}_1^T}{\vdots} \\ \mathbf{q}_n^T \end{bmatrix} \begin{bmatrix} \mathbf{b} \end{bmatrix}$$

and linear combination $\sigma_1 \mathbf{q}_1 + \dots + \sigma_n \mathbf{q}_n = \begin{bmatrix} \mathbf{q}_1 & \cdots & \mathbf{q}_n \end{bmatrix} \begin{bmatrix} \sigma_1 \\ \vdots \\ \sigma_n \end{bmatrix}$



Toward QR factorization (2)

Orthogonalization of columns of matrix A gives us a representation of A as product of

- ▶ matrix with mutually orthogonal columns
- ▶ invertible triangular matrix

$$\left[\begin{array}{c|c|c|c|c} \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 & \cdots & \mathbf{v}_n \end{array} \right] = \left[\begin{array}{c|c|c|c|c} \mathbf{v}_1^* & \mathbf{v}_2^* & \mathbf{v}_3^* & \cdots & \mathbf{v}_n^* \end{array} \right] \left[\begin{array}{cccccc} 1 & \alpha_{12} & \alpha_{13} & & & \alpha_{1n} \\ & 1 & \alpha_{23} & & & \alpha_{2n} \\ & & 1 & & & \alpha_{3n} \\ & & & \ddots & & \\ & & & & 1 & \alpha_{n-1,n} \\ & & & & & 1 \end{array} \right]$$



Toward QR factorization (3)

Orthogonalization of columns of matrix A gives us a representation of A as product of

- ▶ matrix with mutually orthogonal columns
- ▶ invertible triangular matrix

$$\begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 & \cdots & \mathbf{v}_n \end{bmatrix} = \begin{bmatrix} \mathbf{q}_1 & \mathbf{q}_2 & \mathbf{q}_3 & \cdots & \mathbf{q}_n \end{bmatrix} \begin{bmatrix} \|\mathbf{v}_1^*\| & \beta_{12} & \beta_{13} & \cdots & \beta_{1n} \\ & \|\mathbf{v}_2^*\| & \beta_{23} & \cdots & \beta_{2n} \\ & & \|\mathbf{v}_3^*\| & \cdots & \beta_{3n} \\ & & & \ddots & \\ & & & & \beta_{n-1,n} \\ & & & & \|\mathbf{v}_n^*\| \end{bmatrix}$$

Suppose columns $\mathbf{v}_1, \dots, \mathbf{v}_n$ are linearly independent. Then $\mathbf{v}_1^*, \dots, \mathbf{v}_n^*$ are nonzero.

- ▶ Normalize $\mathbf{v}_1^*, \dots, \mathbf{v}_n^*$ (Matrix is called Q)
- ▶ To compensate, scale the rows of the triangular matrix. (Matrix is R)

The result is the QR factorization.

Q is a column-orthogonal matrix and R is an upper-triangular matrix.



Use QR factorization to solve $Ax=b$

QR Solver Algorithm :

Find Q, R such that $A = QR$ and Q is column-orthogonal and R is triangular

Compute vector $\mathbf{c} = Q^T \mathbf{b}$

Solve $R \mathbf{x} = \mathbf{c}$ using backward substitution, and return the solution.

Why is this correct?

- ▶ Let $\hat{\mathbf{x}}$ be the solution returned by the algorithm.
- ▶ We have $R\hat{\mathbf{x}} = Q^T \mathbf{b}$
- ▶ Multiply both sides by Q : $Q(R\hat{\mathbf{x}}) = Q(Q^T \mathbf{b})$
- ▶ Use associativity: $(QR)\hat{\mathbf{x}} = (QQ^T)\mathbf{b}$
- ▶ Substitute A for QR : $A\hat{\mathbf{x}} = (QQ^T)\mathbf{b}$
- ▶ Since Q and Q^T are inverses, we know QQ^T is identity matrix: $A\hat{\mathbf{x}} = \mathbf{1}\mathbf{b}$

Thus $A\hat{\mathbf{x}} = \mathbf{b}$.



solving $A\mathbf{x} = \mathbf{b}$

What if columns of A are not independent? Let $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4$ be columns of A .

Suppose $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4$ are linearly dependent.

Then there is a basis consisting of a subset, say $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_4$

$$\left\{ \left[\begin{array}{c|c|c|c} \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 & \mathbf{v}_4 \end{array} \right] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} : x_1, x_2, x_3, x_4 \in \mathbb{R} \right\} =$$

$$\left\{ \left[\begin{array}{c|c|c} \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_4 \end{array} \right] \begin{bmatrix} x_1 \\ x_2 \\ x_4 \end{bmatrix} : x_1, x_2, x_4 \in \mathbb{R} \right\}$$

Therefore: if there is a solution to $A\mathbf{x} = \mathbf{b}$ then there is a solution to $A'\mathbf{x}' = \mathbf{b}$ where columns of A' are a subset basis of columns of A (and $\mathbf{x}'$ consists of corresponding variables).



The least square problem

Remark: There might not be a solution:

- ▶ Define $f : \mathbb{R}^n \longrightarrow \mathbb{R}^m$ by $f(\mathbf{x}) = A\mathbf{x}$
- ▶ Dimension of $\text{Im } f$ is n
- ▶ Dimension of co-domain is m .
- ▶ Thus f is not onto.

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 6 & 7 & 8 & 9 & 10 \\ 11 & 12 & 13 & 14 & 15 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \mathbf{b}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \\ 10 & 11 & 12 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \mathbf{b}$$

Goal: An algorithm that, given equation $A\mathbf{x} = \mathbf{b}$, where columns are linearly independent, finds the vector $\hat{\mathbf{x}}$ minimizing $\|\mathbf{b} - A\hat{\mathbf{x}}\|$.

Solution: Same algorithm as we used for square A



The least square solution

Recall...

High-Dimensional Fire Engine Lemma: The point in a vector space $\mathcal{V}$ closest to $\mathbf{b}$ is $\mathbf{b}^{\parallel \mathcal{V}}$ and the distance is $\|\mathbf{b}^{\perp \mathcal{V}}\|$.

Given equation $A\mathbf{x} = \mathbf{b}$, let $\mathcal{V}$ be the column space of A .

We need to show that the QR Solver Algorithm returns the vector $\hat{\mathbf{x}}$ such that $A\hat{\mathbf{x}} = \mathbf{b}^{\parallel \mathcal{V}}$.



The representation of $\mathbf{b}^{\parallel}$ in terms of columns of Q

Let Q be a column-orthogonal matrix. Let $\mathbf{b}$ be a vector, and write $\mathbf{b} = \mathbf{b}^{\parallel} + \mathbf{b}^{\perp}$ where $\mathbf{b}^{\parallel}$ is projection of $\mathbf{b}$ onto $\text{Col } Q$ and $\mathbf{b}^{\perp}$ is projection orthogonal to $\text{Col } Q$.

Let $\mathbf{u}$ be the coordinate representation of $\mathbf{b}^{\parallel}$ in terms of columns of Q .

By linear-combinations definition of matrix-vector multiplication,

$$\begin{bmatrix} \mathbf{b}^{\parallel} \end{bmatrix} = \begin{bmatrix} Q \end{bmatrix} \begin{bmatrix} \mathbf{u} \end{bmatrix}$$

Multiply both sides on the left by Q^T :

$$\begin{bmatrix} Q^T \end{bmatrix} \begin{bmatrix} \mathbf{b}^{\parallel} \end{bmatrix} = \begin{bmatrix} Q^T \end{bmatrix} \begin{bmatrix} Q \end{bmatrix} \begin{bmatrix} \mathbf{u} \end{bmatrix}$$



QR solver algorithm for $Ax=b$

Summary:

► $QQ^T \mathbf{b} = \mathbf{b}^{\parallel}$

Proposed algorithm:

Find Q, R such that $A = QR$ and Q is column-orthogonal and R is triangular
Compute vector $\mathbf{c} = Q^T \mathbf{b}$
Solve $R\mathbf{x} = \mathbf{c}$ using backward substitution, and return the solution $\hat{\mathbf{x}}$.

Goal: To show that the solution $\hat{\mathbf{x}}$ returned is the vector that minimizes $\|\mathbf{b} - A\hat{\mathbf{x}}\|$

Every vector of the form $A\mathbf{x}$ is in $\text{Col } A (= \text{Col } Q)$

By the High-Dimensional Fire Engine Lemma, the vector in $\text{Col } A$ closest to $\mathbf{b}$ is $\mathbf{b}^{\parallel}$, the projection of $\mathbf{b}$ onto $\text{Col } A$.

Solution $\hat{\mathbf{x}}$ satisfies $R\hat{\mathbf{x}} = Q^T \mathbf{b}$

Multiply by Q : $QR\hat{\mathbf{x}} = QQ^T \mathbf{b}$

Therefore $A\hat{\mathbf{x}} = \mathbf{b}^{\parallel}$



Application of the least square: regression (1)

Finding the line that best fits some two-dimensional data.

age	brain mass
45	4 lbs.
55	3.8
65	3.75
75	3.5
85	3.3

Let $f(x)$ be the function that predicts brain mass for someone of age x .

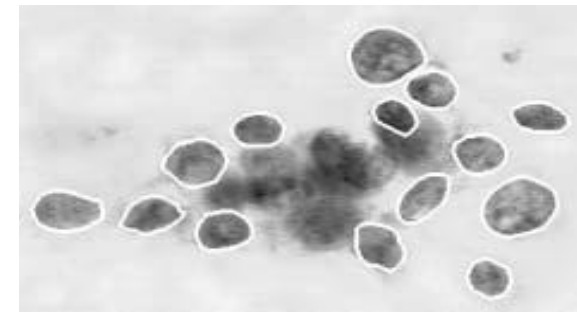
Hypothesis: after age 45, brain mass decreases linearly with age, i.e. that $f(x) = mx + b$ for some numbers m, b .

$$\begin{bmatrix} x_1 & 1 \\ x_2 & 1 \\ x_3 & 1 \\ x_4 & 1 \\ x_5 & 1 \end{bmatrix} \begin{bmatrix} m \\ b \end{bmatrix} \approx \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \end{bmatrix}$$

The dot-product of row i with the vector $[m, b]$ is $mx_i + b$, i.e. the value predicted by $f(x) = mx + b$ for the i^{th} observation.

Therefore, the vector of predictions is $A \begin{bmatrix} m \\ b \end{bmatrix}$.

The vector of differences between predictions and observed values is $A \begin{bmatrix} m \\ b \end{bmatrix} - \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \end{bmatrix}$,



Application of the least square: regression (2)

Recall: breast-cancer machine-learning lab

Input: vectors $\mathbf{a}_1, \dots, \mathbf{a}_m$ giving features of specimen, values $b_1, \dots, b_m$ specifying +1 (malignant) or -1 (benign)

Informal goal: Find vector $\mathbf{w}$ such that sign of $\mathbf{a}_i \cdot \mathbf{w}$ predicts sign of b_i

Formal goal: Find vector $\mathbf{w}$ to minimize sum of squared errors $(b_1 - \mathbf{a}_1 \cdot \mathbf{w})^2 + \dots + (b_m - \mathbf{a}_m \cdot \mathbf{w})^2$

Approach: Gradient descent

Results: Took a few minutes to get a solution with error rate around 7% **Can we do better with least squares?**

Goal: Find the vector $\mathbf{w}$ that minimizes $(\mathbf{b}[1] - \mathbf{a}_1 \cdot \mathbf{w})^2 + \dots + (\mathbf{b}[m] - \mathbf{a}_m \cdot \mathbf{w})^2$

Equivalent: Find the vector $\mathbf{w}$ that minimizes $\left\| \begin{bmatrix} \mathbf{b} \end{bmatrix} - \begin{bmatrix} \mathbf{a}_1 \\ \vdots \\ \mathbf{a}_m \end{bmatrix} \begin{bmatrix} \mathbf{x} \end{bmatrix} \right\|^2$



Lecture 19 - Recap

Previous

orthogonalization

Orthogonal complement

QR factorization

Solving $Ax = b$

Least square regression

Previous
Tools

orthogonalize

Concept & Definition



Thank You For Your Attention!





Applied exercise (1)

Problem: Find a basis for the null space of A using the orthogonal complement.

$$A = \begin{bmatrix} -4 & -1 & -3 & -2 \\ 0 & 4 & 0 & -1 \end{bmatrix}$$



Applied exercise (2)

Problem: Find the QR decomposition of A:

$$A = \begin{bmatrix} 1 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$